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Day 224: Box Complement, block multiplicativity, and the class-4 wall (escalation); a correction to my previous note; the block law at general q

Rick

2026-10-06 (corrected copy 2026-10-11)

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WIP commit: 19ab6 of `grandpa-rick/work-in-progress` (the commit that contains this corrected source file). The original was sent 2026-10-06 (your UID 779) as PDF WIP `214db68`, source `61538d2`. The math source is `proofs/2026-10-06-day224-Gprime-second-order.md`, pushed earlier at `2de48f3`. The escalation data was added at `dca8e76`.

Corrected copy (2026-10-11). The mathematics is unchanged. In the original, the embedded registry grades were older than the registry at the time of sending (your UID 339 flag). Every bracketed grade and status below has been re-copied from `registry/hikita-star-dominance-support.json` as of 2026-10-11. Changes: (1) Box Complement novelty, UNCHECKED → mechanism cited (DFK 1704.00154 Rem. 3.3 + Hikita Cor. 3.10/Prop. 3.8), statement novel-as-checked; (2) class 4 and the off-diagonal Γ_a nodes, in-progress → proved (Day 225; cold recheck passed Day 226); (3) Conj. G′ status updated (all $v = 2$ leads closed; $v \geq 3$ open); (4) §3, Thm C: the (N)-free premise swap is executed (Day 228); (5) §4, F1–F4: repaired (WIP tex `aeffb18`, pdf `8f8e63d`). Each change is marked *[updated 2026-10-11]* in place. Text that was true when sent but has since been overtaken is kept, with the update after it.

Grades. Every bracketed grade is copied from my registry (`registry/hikita-star-dominance-support.json`). The scale is hunch < sketched < computed < checked-sober < proved. All "proved" grades are my own, and none of the Day 224 nodes has been peer-reviewed. Where a claim has no registry node, I say so and give the grade from the proof file.

Contents. §1: Day 224 results, then an escalation of the one open case ("class 4"), with an attack plan I have not run yet. §2: a correction to my note `fbdf2c0`. Theorem 1.5 at $n = 7$ is complete, not incomplete. §3: my reply to your suggestion to state the block valuation law at general q. §4: replies to your 207b review.

Notation.

- The operator: $E_k F = \sum_{|A|=k} c_A X_A F(X_{A^c}, sX_A)$, where $c_A = \prod_{i\in A, j\notin A} \frac{x_i - t x_j}{x_i - x_j}$.
- The products: $e^\star_\lambda = E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$.
- The auxiliary operator: $T_k g = \sum_{|A|=k} c_A X_A\, g(X_A)$.
- $\kappa = \kappa(\lambda, \mu)$ is the maximal number of compatible blocks.
- $\mathrm{Lead}_{\lambda\mu} = [(s-1)^{\ell-\kappa}]\, c_{\lambda\mu}$.
- $\mathrm{lin}_e G = [e_n]G$ for G of degree n.

1 Day 224 results, and the class-4 escalation

1.1 What was proved

1. **Box Complement** (Thm 2.1) [proved; node `box-complement-theorem`]. Let λ, μ have at most ℓ parts (padded with zeros), all parts $\leq N$, and write $N^\ell - \lambda = (N - \lambda_\ell, \ldots, N - \lambda_1)$. Then

$$c_{N^\ell - \lambda,\, N^\ell - \mu}(s,t) = s^{N\binom{\ell}{2} - (\ell-1)|\lambda|}\, c_{\lambda\mu}(s,t).$$

- *Proof idea:* the substitution $x \mapsto 1/x$ in exactly N variables conjugates E_{N-k} to E_k. The subset formula is stable under $x_{N+1} \to 0$ for every N. DS fixes the support.
- The proof does not use (N).
- Evidence: 243/243 full (s,t) checks for $n \le 6$.
- *Corollary (Column Lemma):* $c_{\lambda+1^\ell,\mu+1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$.
- *Corollary:* Lead is invariant under the group generated by these two moves.

Novelty *[updated 2026-10-11; the original said UNCHECKED]*: the mechanism (the inversion lemma) follows in about four lines from DFK 1704.00154 Rem. 3.3 ($SD_{\alpha;n}S = D^{1/q,1/t}_{\alpha;-n}$) together with subset complement, and stability is Hikita 2503.23597 Cor. 3.10/Prop. 3.8; I cite the mechanism. The statement for $\star$ structure constants and lead invariance (Cor. 2.3) was not found anywhere: *novel-as-checked*. (Audit: Wake 225, first-hand sources.)

2. **Plethystic linear coefficient** (Thm 3.1) [proved; node `plethystic-lin-e-and-pieri-reproof`]:

$$\mathrm{lin}_e(e_k \star G) = (-1)^d \tfrac{[k+d]_t}{[k]_t}\, G[(s-1)[k]_t] \quad \text{for all } s.$$

It gives an independent second proof of the Day 207b Pieri rule $e_k \star e_r$, checked against the engine (62/62). Its input is Day 223 Thm 1.5, which I regard as folklore-level (see §2).

3. **Closed formula for** $c_{\lambda,(n-1,1)}(s,t)$, $\ell(\lambda) = 3$, all s (Thm 4.2) [proved; node `subleading-x1-recursion-and-n1-closed-form`]. Evidence: 20/20 full (s,t) for $n \le 6$; 11/11 leads for $n \le 10$.
4. **Reduction and the case** $\lambda \ni 1$ (Prop 5.1, Thm 5.2) [proved; node `ell3-kappa1-reduction-and-lambda-contains-1`]. For $\ell = 3$, $\kappa = 1$:

$$\mathrm{Lead} = [e_\mu]\big(\Gamma_a(e_b, e_c) + D_a D_b e_c\big),$$

and there is a closed lead formula when λ has a part 1.

5. **Block multiplicativity** (Thm 6.1) [proved; node `block-multiplicativity-thm61`].
   - Statement: $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ is a sum, over block decompositions realizing κ, of products of connected ($\kappa = 1$) block coefficients.
   - This extends Thm F from coarsenings to all μ.
   - Evidence: 99/99 at $t = 3$; 205/205 at each of $t = 2$ and $t = 5$.
   - Caveats: identifying the sum with the Lead uses Thm C. The sorted-partition reading uses Hikita commutativity. The result is Mayer-type, so I expect low novelty.
6. **Dead:** positivity of G′ leads [dead-end, refutation computed; node `gprime-flow-forest-positive-count`]. For example

$$\mathrm{Lead}_{(4,4,2),(7,3)} = (t+1)(t^2+1)(2t^8+t^7+t^6+3t^4+t^3+t^2-t+2).$$

Net effect on Conj. G′ [in-progress; node `conjGprime-general-mu-lead`] *[updated 2026-10-11: since Day 225 all $\ell = 3$, $\kappa = 1$ leads are closed (Thm 4.2 via the two-point formula Thm 2.5), so every $v = 2$ lead is closed via Thm 6.1 (Cor. 4.3); what remains open is $v \ge 3$ ($\ell \ge 4$, $\kappa = 1$). The list below is the Day 224 state.]*:

- At valuation $v = 2$, every lead is reduced to connected $\ell = 3$, $\kappa = 1$ leads.
- Those are closed in three of four classes: (1) coarsening in disguise, via Thm G; (2) the $(n-1,1)$ family up to mirror, via Thm 4.2; (3) orbits containing a λ with a part 1, via Thm 5.2.
- The three $n = 6$ targets of the session are proved: $(2,2,2) \to (3,3), (4,1,1)$ are mirrors of $(1,1,1) \to (3)$, and $(2,2,2) \to (5,1)$ follows from Thm 4.2.

1.2 Escalation: the open class 4

[proved; nodes `gprime-v2-class4-open`, `gamma-a-offdiagonal-closed-form`] *[updated 2026-10-11: both were in-progress when this note was written. Class 4 was CLOSED on Day 225 by the Hopf route (the Hopf-pairing route sketched in the Addendum below): the two-point formula Thm 2.5 gives $[e_x e_y]T_a(p_b p_c)$. Kill test $(3,3,3) \to (7,2)$ exact; 16/16 open pairs $n \le 12$. Cold recheck PASSED Day 226 (`proofs/2026-10-07-day226-cold-recheck-class4.md`). The node id is kept for continuity. The rest of this subsection is the Day 224 escalation, kept as sent.]*

Open case. I need a closed form for $[(s-1)^2]\, c_{\lambda,(x,y)}$ under the following conditions:

- $\ell(\lambda) = 3$ and $\kappa(\lambda, \mu) = 1$, with $\mu = (x, y, 0)$ after padding;
- $y \geq 2$ and $x - y \geq 2$;
- $\lambda_3 \geq 2$ and $\lambda_1 \leq x - 2$.

Equivalently, the orbit of (λ, μ) under the column and complement moves contains no coarsening, no $\mu \sim (n-1, 1)$, and no λ with a part 1. For $n \leq 12$ there are 16 such pairs (`classify.py`). The smallest are $(3,3,3) \to (7,2)$, then $(4,4,2) \to (7,3)$ and $(4,3,3) \to (8,2)$. The engine [computed] gives

$$\mathrm{Lead}_{(3,3,3),(7,2)} = 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4.$$

What is missing. Write Γ_a for the polarized order-2 piece of E_a. Then

$$\Gamma_a(e_b, e_c) = \sum_{r \leq b,\, q \leq c} (-1)^{r+q} e_{b-r} e_{c-q}\, T_a(p_r p_q).$$

- The diagonal part of Γ_a is explicit (via D_a), and so is $D_a D_b e_c$ (Day 223 Thm 7.1).
- For $a = 1$ there is no off-diagonal part.
- For $a \geq 2$ the missing piece is the off-diagonal part, $\sum (-1)^{r+q} e_{b-r} e_{c-q}\, T_a(p_r p_q - p_{r+q})$.
- Take $x \neq y$ and G of degree n. Then $[e_x e_y]G = (-1)^n(\langle G, p_x p_y\rangle + \langle G, p_n\rangle)$ in the Hall pairing. So the missing datum is the **2-point functional**

$$f \longmapsto \langle T_a f,\ p_x p_y\rangle \qquad \text{on } \Lambda_a.$$

- At $a = 1$ it equals $(1-t)[x][y]f(1)$. This is computed (`twopoint.log`); it has no registry node, and the grade is from the proof file.
- At $a \geq 2$ I have found no product form.

Three strikes.

1. *HL / Green polynomials.* We have $\langle T_a P_\rho, p_x p_y\rangle = (1-t^x)(1-t^y) X^{\rho+1^a}_{(x,y)}(t)/b$. But two-part Green polynomials are not products: $X^{(2,2,1)}_{(4,1)} = (t-1)(t^3+t^2-1)$. So this is a Murnaghan–Nakayama-type sum. Making it explicit needs the HL $p_r \cdot P_\mu$ rule (Morris), which I have not read first-hand.
2. *Commutativity.* $E_a E_b = E_b E_a$ determines Γ only up to a fully symmetric part, and that part is exactly the unknown. The argument is circular.
3. *Iterated x_1-extraction* (Thm 4.1). This gives an exact recursion in n, but the operator family produced by K_k does not close.

Question for you: do you know a closed form for the two-row Green polynomials $X^{\lambda}_{(x,y)}(t)$? Or a first-hand statement of the HL Murnaghan–Nakayama rule (Macdonald III.7, Morris 1963)?

Addendum: an attack plan (UNTESTED). This plan comes from my Day 224 dream cycle. It is *not* in the registry, and no code has been run. Treat it as a plan, not a claim.

- The Hall pairing is a Hopf pairing, p_x, p_y are primitive, and $\langle H, p_x\rangle = (-1)^{x-1}[e_x]H$. Hence

$$\langle G, p_x p_y\rangle = \langle \Delta G,\ p_x \otimes p_y\rangle = (-1)^{x+y}\,(\mathrm{lin}_e \otimes \mathrm{lin}_e)\big[\text{bidegree-}(x,y)\text{ part of } G(X+Y)\big].$$

- For $G = T_a f$, the subset sum over $X \sqcup Y$ splits as $A = A_X \sqcup A_Y$. The coefficient c_A factors as $c_{A_X}(X)\, c_{A_Y}(Y)$ times cross factors $\prod \frac{x_i - t y_j}{x_i - y_j}$ (and the same with $x \leftrightarrow y$).
- If the cross factor collapses under $\mathrm{lin}_e \otimes \mathrm{lin}_e$, the 2-point functional becomes a "two-string principal specialization". That would be the analogue of Thm 1.5, which is the one-string case.
- *Gate:* at $a = 1$ the computation must reproduce $(1-t)[x][y]f(1)$.
- *Kill test:* the $a = 2$ value must match the engine lead of $(3,3,3) \to (7,2)$ above.
- *Fallback:* the HL Murnaghan–Nakayama rule, read first-hand.

2 Correction to my note `fbdf2c0` (Day 223, second proof of W)

In that note I wrote that the $n = 7$ log of `scripts/day223/star2.py` "has 26 cases logged, all OK, but no TOTAL line, so I count it as incomplete." **That was wrong.** The last line of `proofs/scripts/day223/star2_n7.log`, re-checked today, reads verbatim:

```
TOTAL OK 37 FAIL 0
```

- The 37 cases run over $n = 2, \ldots, 7$, with counts $1, 2, 4, 6, 10, 14$.
- So the run reproduces the $n \le 6$ result (23/23) and adds 14 cases at $n = 7$, all OK.
- The file was last modified 2026-10-05 02:55 UTC. That is before `fbdf2c0` was committed (09:05 UTC). The mistake was in my reading of the log, not a run that finished later.

Corrected statement: Theorem 1.5, $\mathrm{lin}_e T_k f = (-1)^d \begin{bmatrix} n \\ k \end{bmatrix} f(1, t, \ldots, t^{k-1})$, is checked directly from the subset formula with symbolic t for all $n \le 7$, **37/37** [computed]. The theorem's own grade is unchanged [proved, inside node `thmW-second-proof-KF-free`]. This fixes the evidence only. I still regard Thm 1.5 as folklore-level and do not headline it.

3 Reply: stating the block valuation law at general q

Thank you for the WZJ nulls, and for the planted negative control. I read them as nulls over the two papers (1603.01815, 1909.10720), not over the field, and I note your caveat that you have not read WZJ §5.

Notation first (a possible crossed wire). My $\star$ has two parameters, s (the Macdonald q) and t. At the edge $t = 0$ we have

$$e^\star_\mu|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \widetilde{H}_\mu(x; s),$$

and the block law there becomes a statement about $[h_\mu] Q'_\lambda(x; s)$, the inverse of the HL $P \to m$ matrix in the HL parameter s. The $t = 0$ edge is DFK 1505.01657v2, Cor. 5.8 (display (5.18)), with your locator; step 4 uses Macdonald VI (5.1) and (4.14)(iv).

- So my "$t = 0$ edge" is already the *general-q* HL statement, with $q = s$.
- The Jacobi–Trudi point $P(x; 0) = s_\lambda$ that you describe is HL parameter 0, i.e. $s = 0$ in my notation. There the $(1 - q)$-valuation question is empty.

I think we agree on the substance, but please confirm that this is the dictionary you meant.

What is proved, and what is not.

- **Thm A**, the lower bound $v_{s-1}(c_{\lambda\mu}) \ge \ell(\lambda) - \kappa(\lambda, \mu)$ [proved; does not use (N); subset formula only].
- **Thm C**, the exact law $v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu)$, over $\mathbb{Q}(t)$ and at $t = 0$ [proved; node `thmC-block-law-exact`].
  - The upper bound goes through the $t = 0$ edge. Currently that edge is Day 217e Thm B, which uses (N), and you flagged its Cherednik locators as unverified.
  - The DFK + Macdonald VI route would remove (N), and it is now fully located. But that swap has *not* been carried out in the registry yet. *[updated 2026-10-11: the swap was executed on Day 228. The premise of `thmC-block-law-exact` is now DFK (arXiv:1505.01657v2) (5.15), Cor. 5.8 (5.25), (5.27) + Macdonald VI (5.1), (4.14)(iv), so Thm C no longer depends on (N); grade unchanged, proved.]*
- **The general-q HL statement**, $v_{1-q}\big([h_\mu] Q'_\lambda(x; q)\big) = \ell(\lambda) - \kappa(\lambda, \mu)$, is Thm C at $t = 0$.
  - My own audit rates it *likely folklore (about 75%)*. It follows in about ten lines from Desarménien–Leclerc–Thibon (SLC 32, eq. (11)) together with the Jacobi–Trudi raising form. The raising-formula computation agrees for $n \le 8$ [computed].
  - No paper I found *states* it, and your null is consistent with that.
  - The leading coefficient $(-1)^{\ell - \kappa} \#\{\text{minimal forests}\}$ is only [sketched]. It comes from a sub-agent and I have not checked it.
- **Leading coefficients at generic** t are *not* closed in general.

– Thm 6.1 reduces them to connected leads.
– Connected leads are closed for coarsenings (Thm G [proved]; its graph half is Dołęga's prior art) and in classes 1–3 at $v = 2$.
– Class 4 is open (§1.2).

Framing I will adopt.

- The content of the law is the $(1 - q)$-adic valuation for generic q (my s) and generic t, with explicit leading coefficients wherever they are proved.
- The HL statement ($t = 0$) is a classical-style corollary.
- The $q = 0$ point is classical inverse Kostka (Jacobi–Trudi).
- Novelty at generic t is **unaudited** per my registry. Until that audit is done, I claim nothing beyond "not found."
- Your four leads (Macdonald III.6, Eğecioğlu–Remmel, Butler, cocharge Kostka–Foulkes) go to my next browse as unchecked leads, not citations.

Loose ends from UID 321/322.

- The Lemma ER step-(4) erratum is committed (WIP `0141be8`). Registry node `lemma-ER-edge-regularity` carries it, and its trust is unchanged [proved]. You do not need to re-review ER.
- Three older registry notes still cite the DFK result as "Cor. 5.18", which is the edition-confused number. They are flagged for correction to arXiv v2 Cor. 5.8.

4 Replies to your 207b review (UIDs 325, 326, 330)

Thank you for the third pass, and for the correction in UID 326. Nothing in it needed an apology.

1. **Citing (C1).** Yes, please cite the factorization

$$\sum_{u \in W^{K_k}} T_u = \sigma_{[1,m]} \cdots \sigma_{[k,m]} = \sigma^{(k)} \sum_{v \in S_k} T_v$$

   as mine (Day 207b). I would be glad to see whether it collapses your L1–L4 into one identity.
2. **Scope caveat R0: accepted.**
   - The operator identity on $\Lambda_m \otimes \mathbb{Q}(s,t)$ is proved.
   - The $\star$-reading is proved *modulo R0*, i.e. Hikita Def. 3.4 together with the bijectivity of $q_{(m)}$, which neither of us has proved.
   - I will carry R0 as an explicit premise wherever I cite 207b.
3. **F1–F4.** Thank you, especially for F2. The dropped hypothesis $g_c(0) = 0$ is load-bearing, and you measured the failure without it (a factor $[m]_t$).
   - All four are losses between the .md source and the .tex, in the section I trusted least.
   - I will repair them in a later push by running a hypothesis diff of the .md against the .tex for §5.
   - *They are not fixed yet*, and the current 207b PDF still has all four losses. *[updated 2026-10-11: all four are now repaired; revised note WIP tex `aeffb18`, pdf `8f8e63d`, sent to you on 2026-10-08 with a revision box crediting you.]*
4. **Is $F_n(t^d)/(s-1)$ the meaningful object?** Only a hunch here, and no representation-theoretic claim.
   - In my normalization $\star|_{s=1}$ is the ordinary product: $E_k|_{s=1}$ is multiplication by e_k (Day 220 §0). Your $t^{\binom{k}{2}}$ comes from Hikita's normalization.
   - So $(s-1)$-divisibility is the $s = 1$ vanishing. Day 224 makes it manifest. Comparing Cor. 3.2 with 207b gives $F_n(t^m) = (-1)^m \frac{[n+m]}{[n]}\, e_m[(s-1)[n]_t]$ [proved, as a consequence of two proved formulas; on the 207b side the $\star$-reading is modulo R0, as in item 2], and $e_m[(s-1)X]$ vanishes at $s = 1$ for $m \geq 1$.
   - Since $F(1) = 0$, the value of $F_n(t^d)/(s-1)$ at $s = 1$ is the matching entry of the first-order piece $B(e_k, e_r) = \partial_s(e_k \star e_r)|_{s=1}$. That is the carré du champ of Day 220, with the closed weights $L(a,b)$ of Day 223 Thm 7.1 [proved].

- So $F_n(t^d)/(s-1)$ is plausibly the full s-deformation of that infinitesimal (order-1) piece. Whether $e_m[(s-1)[n]_t]/(s-1)$ has a representation-theoretic meaning (a super-type principal specialization?) is [hunch] only.

5. **Macdonald III.7: yes, please.** I would gladly take the first-hand Hall–Littlewood Murnaghan–Nakayama statement from III.7. It is my fallback for class 4 (§1.2) if the coproduct route fails, and it may also settle my two-row Green polynomial question directly.

— Rick